

BRAC UNIVERSITY
CSE331 : Automata and Computability
Assignment 3

1. Use the pumping lemma to show that the following languages are not regular:

A. $L(M) \rightarrow \{0^{2^n}1^n \mid n \geq 0\}$, where $\Sigma = \{0, 1\}$.

See example 4 of the provided video resource on the Pumping Lemma for RL.

B. $L(M) \rightarrow \{0^n1^m0^{f(n,m)} \mid n, m \geq 0\}$, where $f(n, m) = n * m$ and $\Sigma = \{0, 1\}$.

See example 7 of the provided video resource on the Pumping Lemma for RL.

C. $L(M) \rightarrow \{0^n1^m \mid n > m\}$, where $\Sigma = \{0, 1\}$.

See example 3 of the provided video resource on the Pumping Lemma for RL.

D. $L(M) \rightarrow \{0^{n^2} \mid n \geq 0\}$, where $\Sigma = \{0, 1\}$.

See example 10 of the provided video resource on the Pumping Lemma for RL.

E. $L(M) \rightarrow \{0^n \mid n \text{ is a prime number}\}$

See example 10 of the provided video resource on the Pumping Lemma for RL.

F. $L(M) \rightarrow \{w \text{ is not a palindrome}\}$, where $\Sigma = \{0, 1\}$.

See example 18 of the provided video resource on the Pumping Lemma for RL.

G. $L(M) \rightarrow \{0^i1^j2^k \mid i, j, k \geq 0 \text{ and if } i = 1 \text{ then } j = k, \text{ otherwise } j \neq k\}$

Solution. Assume for the sake of contradiction that L is regular. Since it's regular, it has a pumping length, say p . Now we pick the string

$$w = 01^p2^p \in L.$$

Then $|w| = 2p + 1 \geq p$. So w can be split into $w = xyz$ such that

- (i) $|y| > 0$,
- (ii) $|xy| \leq p$, and
- (iii) $xy^iz \in L$ for every $i \geq 0$.

We have **three cases**:

1. $x = \varepsilon, y = 0$.

In this case,

$$xy^2z = 001^p2^p = 0^21^p2^p \notin L,$$

because here $i \neq 1$, but $j = k$.

2. $x = \varepsilon, y = 011 \dots 1$.

In this case,

$$xy^2z = 011 \dots 1 \ 011 \dots 1 \ 11 \dots 122 \dots 2 \notin L,$$

because there is a 0 after 1s.

3. $x \neq \varepsilon$.

In this case, x contains a 0. Since $|xy| \leq p$, and after the first 0, the next p symbols are all 1s, then y consists of 1s only.

$$xy^2z = 01^{p+|y|}2^p \notin L,$$

since $p + |y| \neq p$.

Thus we get a contradiction in all cases. Therefore, L can't be regular. ■

2. Write a CFG for the following CFL:

- A. $L(M) \rightarrow \{0^n 1^m \mid n, m \geq 0 \text{ and } 2n = 3m\}$, where $\Sigma = \{0, 1\}$
 $S \rightarrow 000S11 \mid \varepsilon$
- B. $L(M) \rightarrow \{0^n 1^m 2^m 3^n \mid n, m > 0\}$, where $\Sigma = \{0, 1, 2, 3\}$
 $S \rightarrow 0S3 \mid A \mid \varepsilon$
 $A \rightarrow 1A2 \mid \varepsilon$
- C. $L(M) \rightarrow \{w = 0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and } j < 2i + k\}$, where $\Sigma = \{0, 1, 2\}$
 $S \rightarrow XABY \mid XA \mid BY$
 $A \rightarrow 0A11 \mid \varepsilon$
 $B \rightarrow 1B2 \mid \varepsilon$
 $X \rightarrow 0X \mid 0$
 $Y \rightarrow 2Y \mid 2$
- D. $L(M) \rightarrow \{w_1 \# w_2 \mid \text{the number of } 00 \text{ in } w_1 \text{ is equal to the number of } 11 \text{ in } w_2\}$, where $\Sigma = \{0, 1\}$
 $S \rightarrow 0B1 \mid XS \mid SY \mid Z$
 $X \rightarrow 1X \mid 01X \mid \varepsilon$
 $Y \rightarrow 0Y \mid 01Y \mid \varepsilon$
 $Z \rightarrow 0\# \mid \#1 \mid 0\#1 \mid \#$
 $A \rightarrow 0B \mid 1A$
 $B \rightarrow 0D \mid 1A$
 $C \rightarrow D1 \mid C0$
 $D \rightarrow E1 \mid 0C$
 $E \rightarrow B \mid \#$
- E. $L(M) \rightarrow \{w\#x \mid w^R \text{ is a substring of } x\}$, where $\Sigma = \{0, 1\}$
 $S \rightarrow AB$
 $A \rightarrow 0A0 \mid 1A1 \mid \#B$
 $B \rightarrow 0B \mid 1B \mid \varepsilon$
- F. $L(M) \rightarrow \{w \mid w \text{ is not a palindrome}\}$, where $\Sigma = \{0, 1\}$.
 $S \rightarrow 0S0 \mid 1S1 \mid 0A1 \mid 1A0$
 $A \rightarrow 0A \mid 1A \mid \varepsilon$

3. Using the string 'abaabbba' show that the following grammar is ambiguous:

- $S \rightarrow PbQ$
 $P \rightarrow aP \mid bP \mid a \mid b$
 $Q \rightarrow aQa \mid aQb \mid bQa \mid bQb \mid \varepsilon$

1. $S \Rightarrow PbQ \Rightarrow abQ \Rightarrow abaQa \Rightarrow abaaQba \Rightarrow abaabQbba \Rightarrow abaabbba$

2. $S \Rightarrow PbQ \Rightarrow aPbQ \Rightarrow abPbQ \Rightarrow abaPbQ \Rightarrow abaaPbQ \Rightarrow abaabbQ \Rightarrow abaabbbQa \Rightarrow abaabbba$

As we have 2 LMDs for the string 'abaabbba', the grammar is ambiguous.